MIL-L-11848C
26 February 1976
SUPERSEDING
MIL-L-11848B
24 April 1969

MILITARY SPECIFICATION

LIGHTS, POWER TOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of portable, adjustable, incandescent, electric lights suitable for mounting on lathes or other power tools.

1.2 Classification. The lights shall be of the following types, as specified (see 6.2):

Type I - Flexible adjustment.
Type II - Floating adjustment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

J-C-175	- Cable Assembly, Power, Electrical (3-Wire, 3 Prong, Grounding Plug Connector, for 125-Volt Equipment).
TT-E-485	- Enamel, Semi-Gloss, Rust-Inhibiting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	- Cushioning Material, Cellulosic.

FSC 6230

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Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

Federal

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

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3. REQUIREMENTS

3.1 Description. The lights shall consist essentially of a lampholder, a cable with grounding plug, a reflector, a device to adjust the position of the lamp, and a universal-type mounting base to permit clamping or permanent mounting to either vertical or horizontal components of power tools or work benches.

3.2 First article (preproduction model). The contractor shall furnish one light for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce lights that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4 Lampholder. The lampholder shall have a medium screw-base socket with a molded phenolic or porcelain shell. The socket shall have a rating of 250 volts and 250 watts, and shall incorporate a switch of the push type or key type. The lampholder shall be mounted in and supported by the reflector. The lampholder shall be detachable from the reflector.

3.5 Reflector. The reflector shall totally shield a 60-watt medium screw base incandescent lamp. The reflector shall be attached to the arm of the lampstand.

3.6 Cable and connector plug. The light shall be equipped with an 18 AWG, Type CO-03MOF (3/18) 0330 cable assembly conforming to J-C-175. Length of the cable shall be 9 feet, plus or minus 2 inches, including the connector plug. All metal parts of the light shall be grounded.

3.7 Treatment and painting. All surfaces requiring painting shall be treated and painted in accordance with MIL-T-704, Type A. Where dissimilar metals are used in contact with each other, the surfaces shall be given two coats of paint conforming to TT-E-485 or separated by a layer of vinyl-zinc tape. Dissimilar metals are defined in MIL-T-704. Where welding or brazing is done after painting, the area shall be cleaned and repainted as specified herein.

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3.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.8.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.8.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government, that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.9 Identification marking. The lights shall be identified in accordance with MIL-STD-130.

3.10 Type I, flexible adjustment.

3.10.1 Flexible arm. The flexible arm shall consist of approximately 10 inches of rigid tubing of the same diameter, forming a combined length of 25 inches, plus or minus 2 inches, less the reflector. The rigid tubing end of the flexible arm shall be fastened to the mounting base and the flexible tubing end shall be secured to the reflector assembly. The cable shall be routed within the rigid and flexible arm.

3.10.2 Mounting base. The base shall be a universal type designed for vertical and horizontal mounting, not less than 3 inches wide. The base shall include a removable auxiliary "C" clamp with an opening of not less than 2 inches and a throat depth of not less than 1-3/4 inches.

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3.11 Type II, floating adjustment. The light shall consist of two hinged arms which connect to the base and reflector assembly. The cable shall be routed within the tubular arms. The light shall retain the lampholder and reflector in any desired position under vibrations normally encountered in machine-tool operation.

3.11.1 Arms. Each arm shall be of rigid tubing approximately 15 inches long. The two arms shall be hinged together by means of a friction joint to provide both horizontal and vertical movement of 35 inches (including joint), plus or minus 2 inches. The arms shall be fastened to the reflector assembly and to the mounting base by friction joints. A means shall be provided to allow the reflector assembly to rotate on the fixture arm 180 degrees in either direction.

3.11.2 Mounting base. The mounting base shall be as specified in 3.10.2, except that a stop shall be provided to restrict the light from rotating more than 325 degrees, plus or minus 25 degrees.

3.12 Workmanship. The light shall be fabricated and finished to insure freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, thoroughness of soldering, welding, brazing, painting, and riveting, alinement of parts, and tightness of assembly components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction light shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Test. The preproduction light shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test.

4.4.3.1 Individual. Each light shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The lights shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Parts or components missing.
103. Materials or components not as specified.
104. Misalinement of parts or components.

Minor

201. Identification marking missing or not as specified.
202. Workmanship not as specified.

4.5.2 Operational test. The light shall be equipped with a 60-watt incandescent lamp connected to a power source and operated. Failure of the lamp to light shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

105. Materials and containers not as specified.
106. Cushioning not adequate to prevent movement and damage.
107. Box closure not as specified for Level A or B.
108. Gross weight exceeds 200 pounds for Level A.
109. Gross weight or size exceeds the limitations of the applicable box specification for Level A or B.
110. Strapping not zinc coated for Level A.
111. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging will be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each light shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. Cushioning conforming to PPP-C-843, Type II, Class B shall be used to prevent movement and damage of the light within the box. The box shall be waterproof as specified for waterproofing of slotted-style boxes in accordance with the appendix to the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with FED. STD. No. 356.

5.1.3 Level C. Each light shall be packaged in a close-fitting fiberboard box in a manner to afford protection against deterioration and damage from the contractor to the initial destination. The contractor's standard commercial packaging is acceptable when it fulfills these requirements.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The lights, preserved as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. The gross weight of each box shall not exceed 200 pounds. The box shall be closed and strapped in accordance

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with the appendix of the box specification. Strapping shall conform to QQ-S-781, Class I, Type I or IV, size as applicable unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The lights, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A except the boxes shall be Domestic Type, or the lights shall be packed in boxes conforming to PPP-B-640, Class 2, style optional; or PPP-B-636, V3c. The gross weight or size of each fiberboard box shall not exceed the limitations of the applicable box specification. Fiberboard boxes shall be waterproof sealed as specified in the appendix of the box specification.

5.2.3 Commercial. Commercial packing shall be in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial preservation and packing shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The Type I light is primarily intended for use on power tools where extreme adjustments in position of the light will be required. The Type II light is intended for use on power tools or workbenches where a more fixed position will be retained.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of light required (see 1.2).
(c) Time frame required for submission of preproduction model light (see 3.2).
(d) Preservation and packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

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6.3 Preproduction model light. Any changes or deviations of production lights from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish lights conforming to this specification.

Custodians:

Army - ME
Air Force - 82

Review activity:

DSA - GS

User activity:

Navy - SH

Preparing activity:

Army - ME

Project No. 6230-0219

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-L-11848C Lights, Power Tool

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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